

Reproducing ConInfer: Context-Aware Inference for Training-Free Open-Vocabulary Remote Sensing Segmentation

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Abstract—Open-vocabulary remote sensing segmentation (OVRSS) labels land cover in aerial images from textual queries, without task-specific training. ConInfer (CVPR 2026 Findings) uses several images together to learn spatial context and in this way improves training-free OVRSS. This report reproduces ConInfer on the LoveDA validation set with the authors’ public code on a single free-tier GPU (Google Colab T4). Processing 50 images jointly gives 39.78 mIoU, compared with 39.33 mIoU reported in the paper. We then investigate how the number of jointly processed images affects accuracy: with the same code and weights, mIoU improves from 36.99 (8 images) to 39.02 (25) and 39.78 (50), and all classes benefit. We also record the differences between our environment and the original one and discuss the limitations of the reproduction.

Index Terms—remote sensing, open-vocabulary semantic segmentation, training-free inference, CLIP, DINOv3, reproducibility

I. Introduction

Training datasets for semantic segmentation of remote sensing images are typically densely labelled with a fixed set of classes. An open-vocabulary approach removes this requirement: a vision–language model such as CLIP [1] scores each image region against text descriptions of the classes, so new classes need no new annotations. Training-free variants adapt CLIP’s dense features directly [2], [3], and SegEarth-OV [4] adapts this idea to aerial images. Most of these techniques classify each patch separately, without considering that patches of the same land-cover class look alike both within one image and across a set of images. ConInfer [5] introduces a context-aware inference step that uses self-supervised DINOv3 [6] features to jointly refine the CLIP predictions over a set of images.

This report is based on the project brief of reproducing a recent (2026) computer-vision paper together with the code it has released. Our contributions are:

- a reproduction of ConInfer on the LoveDA benchmark [7] on free-tier hardware, which lands within 0.45 mIoU of the published number;
- a controlled study of the number of images processed together (8, 25, 50), which shows that accuracy depends strongly on this setting: the more images are processed together, the higher the mIoU;

- an account of the changes to the environment that were required to execute the code, so that others can repeat the experiment.

II. Related Work

CLIP-based dense prediction. CLIP [1] is trained with image-level captions, so its patch features are not well suited to dense prediction. MaskCLIP [2] and ClearCLIP [3] modify the final attention layers to obtain usable per-patch features without training.

Open-vocabulary segmentation for remote sensing. SegEarth-OV [4] extends training-free CLIP segmentation to aerial and satellite images, in which objects are small and the viewpoint differs from natural images. It is the key baseline of ConInfer.

Self-supervised visual features. DINOv3 [6] offers dense features that are good for grouping semantically similar regions, and includes a satellite variant (SAT-493M) that ConInfer uses for aerial data.

III. ConInfer in Brief

ConInfer is a training-free inference procedure with two sources of information. A CLIP branch provides, for each image patch i , a probability p_{ik} that the patch belongs to class k (a prior). A DINOv3 branch provides a feature vector $\mathbf{x}_i$ for each patch. For a group of images, these features are modelled by a Gaussian mixture whose components correspond to the classes. The mixture posterior q_{ik} represents how well patch i matches the feature cluster of class k , and carries information from all patches in the group.

The two are combined as

$$z_{ik} \propto p_{ik} q_{ik}, \quad (1)$$

and the refined assignments z_{ik} are used to update the mixture; this is repeated for several iterations (10 in the paper’s setting). The correction is applied after every 50 images, so the number of images processed together is part of the method. The highest-scoring class of a patch is its final label.

TABLE I
LOVEDA VALIDATION mIoU (%). “PAPER” ROWS ARE COPIED FROM TABLE I OF [5]; “OURS” WAS MEASURED IN THIS WORK. SEG-EARTH-OV AND MASKCLIP* WERE NOT RE-RUN BY US.

Method	Source	mIoU
MaskCLIP*	Paper	31.58
SegEarth-OV	Paper	35.79
ConInfer	Paper	39.33
ConInfer (batch 50)	Ours	39.78

IV. Experimental Setup

A. Data and Procedure

We use the validation set of LoveDA [7] (7 classes: background, building, road, water, barren, forest, agricultural), available from the official Zenodo release (1669 images). Only the validation split was converted and used, as the method is training-free. Metrics are mean intersection over union (mIoU), overall accuracy (aAcc) and mean class accuracy (mAcc), as reported by the evaluation script.

B. Implementation

We used the authors’ code¹ with the LoveDA configuration. The settings follow the paper: CLIP ViT-B/16, DINOv3 ViT-L/16 pretrained on SAT-493M, input resized to 448×448 , and no sliding window. The CLIP branch uses the segmentor file from the SegEarth-OV repository.

C. Environment and Deviations

All runs used a Google Colab NVIDIA T4 GPU. To execute the code there we made the following changes, which might account for some differences from the published numbers:

- PyTorch 2.7.1 (CUDA 12.6); mmcv-lite 2.1.0 instead of the full mmcv, with a stub for compiled operators that the evaluation path does not use;
- the DINOv3 SAT-493M weights were downloaded from a third-party Hugging Face repository (MVRL/dinov3_vitl16_sat);
- the paths in the segmentor file were hard-coded; they were changed to Colab paths and num_workers was reduced to 2.

The code is deterministic in our environment: repeating the batch-8 and the batch-50 runs gave identical numbers.

V. Results

A. Reproduction on LoveDA

Table I compares our result with the numbers published in [5]. With a batch size of 50, which corresponds to the group size used in the paper, we obtained 39.78 mIoU, 0.45 higher than the 39.33 reported in the paper.

B. Effect of the Number of Jointly Processed Images

We changed only the batch_size value of the configuration and kept code, weights and data fixed (Table II, Fig. 1).

¹<https://github.com/Dog-Yang/ConInfer>

TABLE II
EFFECT OF THE BATCH SIZE (IMAGES PROCESSED TOGETHER) ON LOVEDA VALIDATION RESULTS (%).

Batch size	aAcc	mIoU	mAcc
8	52.86	36.99	59.82
25	55.43	39.02	60.87
50	56.15	39.78	61.38

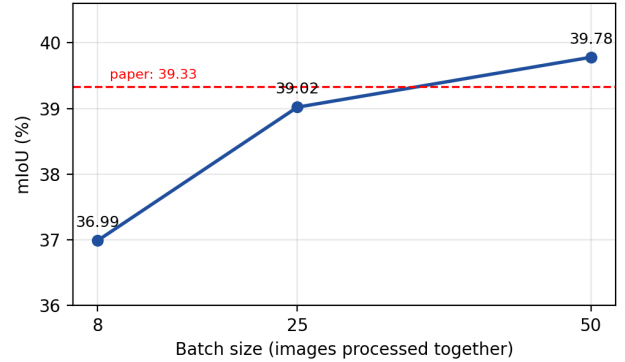


Fig. 1. mIoU on LoveDA validation versus the number of jointly processed images.

mIoU improves monotonically with the group size: from 8 to 25 images it improves by 2.03 points, and from 25 to 50 images by a further 0.76 points. The gain per added image thus decreases, but from three settings we cannot determine whether the curve saturates beyond 50. The trend is in line with the method design: the larger the group, the better the feature distribution of each class can be estimated. This explanation has not been checked by changing the code; it is only our interpretation of the observed trend.

C. Per-Class Results

Table III shows the IoU per class for the smallest and the paper-matching group size. All seven classes improve with batch 50. The largest gains are for agricultural land (+4.31), building (+3.88) and background (+3.34), and the smallest for forest (+1.08) and water (+1.54). The mean of these seven gains equals the 2.79-point mIoU improvement.

D. Computational Cost

As the whole group is processed at once, memory consumption grows with the group size. The logs show 2.1 GB of GPU memory per iteration at batch 8 and 8.3 GB at batch 50, so the 15 GB T4 GPU can process batch 50 but has little room left for larger batches. Evaluation time is about the same for all three settings (around four to five minutes for the 1669 images), because fewer, larger iterations take proportionally longer.

VI. Discussion and Limitations

Despite the deviations from the published environment, our reproduction gives a slightly higher mIoU than the published one. The limitations we recognize are:

TABLE III
PER-CLASS IOU (%) ON LOVE-DA VALIDATION.

Class	Batch 8	Batch 50	Gain
Background	24.55	27.89	+3.34
Building	47.90	51.78	+3.88
Road	36.08	39.03	+2.95
Water	53.70	55.24	+1.54
Barren	18.75	21.19	+2.44
Forest	29.36	30.44	+1.08
Agricultural	48.57	52.88	+4.31

- **One dataset and split.** We only evaluated LoveDA validation; other benchmarks of the paper were not run.
- **Baselines not re-run.** The SegEarth-OV and MaskCLIP* numbers are taken from the paper, so the comparison mixes our hardware and software with the authors’.
- **Environment differences.** The +0.45 difference may be due to the third-party DINOv3 weights, the stubbed `mmcv` operators and the library versions; we did not separate the contribution of each of them.
- **Partial last batch.** 1669 images do not divide evenly into the group sizes, so the last group has only 19 (batch 25 and 50) or 5 (batch 8) images and receives less context.
- **No qualitative analysis.** We report quantitative results only.

VII. Conclusion

We replicated ConInfer, a CVPR 2026 Findings paper on training-free open-vocabulary remote sensing segmentation, using free-tier hardware and achieved 39.78 mIoU, close to the reported 39.33. The accuracy is highly sensitive to the number of images processed together (36.99 to 39.78 mIoU for 8 to 50 images), which is an important parameter to keep when comparing with the published results. Future work includes testing the other benchmarks of the paper, running the SegEarth-OV baseline in the same environment, and testing on other remote sensing data.

Availability

Paper: <https://arxiv.org/abs/2603.29271>. Original code: <https://github.com/Dog-Yang/ConInfer>. Our notebook and results: <https://github.com/mamoonashafique17-stack/ConInfer-reproduction>.

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